Optimized Placement and Pricing Framework for Electric Vehicle Charging Stations

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By

Ankit Mewada : 2021IMT-013

Under the Supervision of

Prof. Aditya Trivedi

and

Dr. Saumya Bhadauria

विश्वजीवनामृतं ज्ञानम्

ABV-INDIAN INSTITUTE OF INFORMATION TECHNOLOGY
AND MANAGEMENT GWALIOR 474015

MADHYA PRADESH, INDIA

DECLARATION

I hereby certify that the work presented in this report/thesis, entitled **"Optimized Placement and Pricing Framework for Electric Vehicle Charging Stations"**, submitted in partial fulfillment of the requirements for the Integrated Post Graduate - Master of Technology in Information Technology, is an authentic record of my own work carried out during the period **July-2025 to December-2025** under the supervision of **Dr. Aditya Trivedi** and **Dr. Saumya Bhadauria**. All references, figures, and tables taken from other sources have been properly cited.

Dated: **Signature of the candidate**

This is to certify that the above statement made by the candidate is correct to the best of our knowledge.

Dated: **Signature of Supervisor: Prof. Aditya Trivedi**

Dated: **Signature of Supervisor Dr. Saumya Bhadauria**

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Ankit Mewada

Abstract

This thesis presents a hybrid optimization framework to enhance the efficiency, scalability, and sustainability of Electric Vehicle Charging Station (EVCS) networks. With the rapid growth of electric mobility, optimizing charging infrastructure has become essential to balance placement, pricing, and grid stability. The proposed dual-stage hybrid model integrates Bender's Decomposition and the Non-Dominated Sorting Genetic Algorithm-II (NSGA-II) to jointly optimize charging station locations and dynamic pricing strategies. Formulated as a multi-objective, mixed-integer, non-linear problem, the framework captures both discrete and continuous decision variables. The Bender's Decomposition-based master problem identifies optimal station sites considering population density, EV demand, and grid accessibility, while the NSGA-II subproblem determines Pareto-optimal pricing strategies balancing affordability, profitability, and demand distribution. Using real-world data from Hefei City, China, the framework achieves a 35% improvement in charging coverage, optimal pricing between 1.0–1.3 ¥/kWh, and significant reductions in waiting times and grid imbalance. The results validate the model's effectiveness and scalability for intelligent, data-driven EVCS network planning, with potential extensions toward renewable energy integration, stochastic pricing, and real-time grid coordination for sustainable urban mobility.

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List of Acronyms

API	Application Programming Interface
BESS	Battery Energy Storage System
BMS	Battery Management System
BIC	Binary Integer Constraint
CAGR	Compound Annual Growth Rate
CAPEX	Capital Expenditure
CSV	Comma Separated Values
CV	Constraint Violation
DEAP	Distributed Evolutionary Algorithms in Python
DER	Distributed Energy Resources
DSS	Distribution Substation System
DTO	Demand–Tariff Optimization
EV	Electric Vehicle
EVCS	Electric Vehicle Charging Station
EVSE	Electric Vehicle Supply Equipment
EVCN	Electric Vehicle Charging Network
FCFS	First-Come First-Served (Queue Discipline)
GA	Genetic Algorithm
GIS	Geographic Information System
GPU	Graphics Processing Unit
ILP	Integer Linear Programming

IoT	Internet of Things
LAT	Load Allocation Table
LMP	Locational Marginal Pricing
M/M/c/K	Finite-Capacity Queueing Model
MoE	Multi-Objective Evaluation
MOO	Multi-Objective Optimization
NSGA-II	Non-Dominated Sorting Genetic Algorithm–II
OPEX	Operational Expenditure
OSM	OpenStreetMap
PV	Photovoltaic (Solar Power)
PDF	Probability Density Function
PCC	Point of Common Coupling
QoS	Quality of Service (Queue Performance)
RES	Renewable Energy Sources
ROI	Return on Investment
SCADA	Supervisory Control and Data Acquisition
SOC	State of Charge (EV Battery)
SP	Subproblem (Benders Decomposition)
SPP	Station Placement Problem
SR	Service Rate (Queueing)
TCO	Total Cost of Ownership
TSP	Tariff Setting Problem (Pricing Submodel)
ULF	Utilization Load Factor
Wq	Average Waiting Time in Queue

1

Introduction

This chapter introduces the motivation and background for developing a data-driven, multi-objective optimization framework for large-scale Electric Vehicle Charging Station (EVCS) planning in rapidly growing urban environments. Section 1.1 provides a general introduction to the research context, emphasizing the increasing importance of optimized EVCS networks for sustainable mobility in cities such as Indore. Section 1.2 outlines the technical background—covering EV adoption trends, spatial demand characteristics, grid constraints, and the rationale for integrating NSGA-II, Benders decomposition, CAGR-based demand forecasting, and queueing models in EVCS planning. Section 1.3 highlights the practical and methodological motivations guiding the work, focusing on the need for scalable, data-driven approaches that balance cost, coverage, service quality, and profitability. Section 1.4 lists the primary objectives of the thesis, including optimal station placement, dynamic pricing optimization, and operational performance evaluation. Finally, Section 1.5 presents the overall layout of the report, summarizing the structure and interconnections of the subsequent chapters.

Abstract

This thesis presents a data-driven, multi-objective optimization framework for the planning and deployment of Electric Vehicle Charging Station (EVCS) networks in rapidly growing urban environments, with Indore city as the case study. As electric vehicle (EV) adoption accelerates, charging infrastructure must be designed to balance investment cost, spatial coverage, user accessibility, grid stability, and economic viability.

The proposed framework integrates a non-dominated sorting genetic algorithm (NSGA-II) with Benders decomposition, coupled with a Compound Annual Growth Rate (CAGR)-based demand model and queueing-theoretic performance evaluation. The NSGA-II component optimizes EVCS site selection across 232 candidate locations to serve demand from 288 zones, while explicitly trading off total cost, demand coverage, average service distance, and annual profit. For each candidate deployment, a Benders decomposition-based pricing module determines site-level charging prices within realistic bounds, maximizing profit under grid and demand constraints. A CAGR model derived from official 2011 and 2021 census data projects population and EV fleet growth up to 2031, and M/M/c/K queueing models estimate waiting times, utilizations, and stability for each selected station.

Applied to Indore, the framework yields an optimal 2021 baseline network with 90 stations, total investment of approximately 50.72 Crore, 100% demand coverage, an average service distance of 0.70 km, and an expected annual profit of 64.55 Lakh. Queue analysis shows average waiting times under one minute during normal hours and around 2.26 minutes during peak hours, with mean utilization around 0.42. Long-term projections indicate that the baseline network must expand by more than a factor of three to

meet 2031 demand, motivating a phased deployment strategy in three stages (2021–2023, 2024–2027, 2028–2031).

The results demonstrate that the proposed hybrid framework is effective and scalable for city-scale EVCS planning. It supports evidence-based decisions on site selection, pricing, and capacity expansion, and can be extended to incorporate renewable integration, stochastic demand modeling, and real-time operational coordination for sustainable urban e-mobility.

1.1 Introduction

This chapter introduces a hybrid optimization framework for the efficient and sustainable deployment of Electric Vehicle Charging Station (EVCS) networks in Indore city. The framework combines multi-objective site-selection, dynamic pricing, growth-aware demand modeling, and queueing-based performance evaluation into a unified planning tool.

Section 1.1.1 provides a high-level overview of the proposed model, which integrates NSGA-II for site selection and Benders decomposition for pricing. Section 1.2 discusses the broader context of EV adoption and charging infrastructure planning in rapidly urbanizing Indian cities. Section 1.3 presents the motivation for developing a data-driven, scalable framework that balances infrastructure cost, accessibility, profitability, and grid stability. Section **??** summarizes the key objectives of the thesis, and Section 1.5 outlines the overall structure of the report. Together, these sections establish the scope, goals, and methodological direction of the proposed EVCS optimization framework.

1.1.1 Overview of the Proposed Framework

The global transition toward sustainable transportation has led to a rapid increase in the adoption of Electric Vehicles (EVs). In India, and particularly in fast-growing cities such as Indore, this trend is reinforced by government incentives, environmental regulations, and rising public awareness. However, large-scale EV uptake is fundamentally

constrained by the availability, accessibility, and reliability of the charging infrastructure. Poorly planned EVCS networks can lead to range anxiety, long waiting times, uneven spatial coverage, and overloading of the power grid.

Designing an optimized EVCS network is a complex, multi-objective task. It involves simultaneously deciding:

- which candidate locations should host charging stations;
- how many chargers and what capacity should be installed at each selected site;
- how charging prices should be set across the network;
- how to ensure that grid constraints and user service requirements are satisfied.

These decisions must be made in the presence of spatially heterogeneous demand, limited grid capacity, and evolving population and EV growth over a multi-year horizon.

To address this challenge, the thesis proposes a hybrid framework with four tightly coupled components:

(i) **NSGA-II based site selection:** A multi-objective NSGA-II optimizer selects EVCS sites from a set of 232 candidates to serve demand from 288 zones and 36 grid nodes. The objective vector includes total investment cost, demand coverage, average service distance, and annual profit.

(ii) **Benders-based pricing optimization:** For each candidate deployment from NSGA-II, a pricing subproblem is solved using Benders decomposition. This subproblem optimizes station-level prices within realistic bounds (e.g., 8–15 INR/kWh), maximizing profit while respecting demand elasticity and operational constraints.

(iii) **CAGR-based demand forecasting:** Population and EV fleet sizes are projected from 2021 to 2031 using a Compound Annual Growth Rate (CAGR) model based on official census data. These projections drive long-term demand scenarios and phased expansion planning.

(iv) **Queueing-theoretic performance evaluation:** M/M/c and M/M/c/K queueing models estimate waiting times, queue lengths, and utilization at each station, ensuring that optimized deployments are not only spatially efficient but also operationally stable.

Applied to Indore, this framework produces a baseline 2021 solution with 90 stations, 100% demand coverage, an average service distance of 0.70 km, and an annual profit of approximately 64.55 Lakh, for a total investment of about 50.72 Crore. These results form the starting point for phased expansion strategies up to 2031.

1.2 Background

The expansion of EV infrastructure in Indian cities requires optimization techniques capable of handling multiple layers of complexity. Unlike conventional fuel stations, EVCSs interact directly with the power distribution network, making their planning and operation depend simultaneously on transportation patterns, demographic trends, and grid constraints.

1.2.1 EV adoption and urban charging needs

India's urban centers are experiencing rapid demographic growth and motorization. For Indore specifically, census data indicate that the population grew from about 1.87 million in 2011 to 3.67 million in 2021, corresponding to a CAGR of roughly 7.04%. Projections suggest that the population may exceed 7.2 million by 2031. Under realistic EV adoption scenarios, the EV fleet is expected to increase by more than a factor of eight over the same period.

Such growth patterns imply that EVCS networks cannot be designed solely for current demand. Instead, they must be robust to substantial increases in both the number of EVs and their spatial distribution. This requires demand modeling that explicitly accounts for long-term population and EV growth, as well as spatial heterogeneity across urban zones.

1.2.2 Limitations of traditional planning approaches

Traditional EVCS planning approaches often rely on:

- single-objective deterministic optimization (e.g., minimizing average distance or total cost),
- simplified grid models that ignore detailed capacity constraints,
- static demand assumptions that do not consider long-term growth.

While these models can provide initial insights, they are insufficient for city-scale planning with multiple, conflicting objectives. In particular, they struggle to:

- capture trade-offs between cost, coverage, service distance, and profit;
- incorporate realistic grid constraints and upgrade costs;
- integrate dynamic pricing strategies and demand elasticity;
- ensure acceptable queueing performance during peak periods.

These limitations motivate the use of multi-objective evolutionary algorithms and decomposition techniques that can explicitly model trade-offs and scale to large urban instances.

1.2.3 NSGA-II, Benders decomposition, and queueing models

NSGA-II is an elitist multi-objective genetic algorithm widely used for problems with conflicting objectives and large, combinatorial search spaces. Its fast non-dominated sorting, crowding distance mechanism, and elitism make it suitable for exploring diverse Pareto-optimal solutions in EVCS site selection.

Benders decomposition is a classical mathematical programming strategy that partitions a large problem into a master problem and one or more subproblems. In this thesis, structural decisions (site selection) are handled by NSGA-II, while pricing optimization

for a given network configuration is treated as a continuous subproblem, solved iteratively and feeding back Benders cuts that refine the feasible region and improve solution quality.

Finally, M/M/c and M/M/c/K queueing models provide analytical expressions for queue lengths, waiting times, and utilization at charging stations. They are used here to ensure that the proposed deployments remain operationally feasible and do not lead to excessive waiting times or unstable behavior (i.e., utilization $\rho \geq 1$) at high-demand sites.

1.3 Motivation

The motivation for this research arises from both practical infrastructure requirements and methodological limitations in existing EVCS planning approaches.

1. Addressing infrastructure inefficiencies

Many emerging EVCS networks suffer from uneven spatial coverage and demand imbalance: some stations remain underutilized, while others experience congestion and long queues. In Indore, strong population growth and evolving urban form further complicate the allocation of stations across zones. A systematic, data-driven optimization framework is needed to ensure that infrastructure is deployed where it is most effective, minimizing waste and improving user experience.

2. Balancing affordability, profitability, and coverage

Charging price is a critical lever for both user adoption and operator revenue. Prices that are too high discourage EV usage and may lead to underutilized infrastructure, whereas prices that are too low can undermine financial sustainability. By integrating Benders-based pricing into the planning model, this thesis seeks to identify charging tariffs that are affordable for users, profitable for operators, and compatible with long-term expansion and maintenance costs.

3. Incorporating real-world constraints and growth

EVCS planning in Indore must account for:

- spatially distributed demand across 288 zones;
- grid capacities and upgrade costs at 36 power nodes;
- land-use and practical siting constraints at 232 candidate locations;
- projected population and EV growth up to 2031.

The proposed framework integrates these constraints explicitly, leading to solutions that are not only theoretically optimal but also realistic and implementable.

4. Ensuring operational feasibility via queueing analysis

Even optimally located stations can fail if they generate long queues and high waiting times. Queueing analysis is therefore essential to verify that station configurations deliver acceptable levels of service. By incorporating M/M/c and M/M/c/K models, the thesis ensures that selected deployments maintain low waiting times and stable utilizations across a range of demand scenarios.

5. Providing a reusable, extensible planning tool

Beyond the specific case of Indore, the methodology is designed to be reproducible and extensible. The combination of open data, modular Python implementation, evolutionary optimization, and analytical queueing models yields a framework that can be adapted to other cities with similar demographic and grid profiles, and extended to incorporate renewable integration, vehicle-to-grid (V2G) capabilities, and real-time control.

1.4 Thesis Objectives

The specific objectives of this thesis are:

(i) To develop a multi-objective optimization model using NSGA-II for the optimal placement of Electric Vehicle Charging Stations (EVCS) in Indore, jointly considering total investment cost, demand coverage, and average service distance.

(ii) To integrate a pricing optimization module based on Benders decomposition that determines station-wise charging tariffs within realistic bounds, maximizing annual profit while respecting grid and demand constraints.

(iii) To evaluate the operational performance of the optimized EVCS network using queueing models (M/M/c and M/M/c/K), quantifying key metrics such as waiting time and charger utilization to ensure service quality and system stability.

1.5 Layout of the Thesis

This report is structured into the following chapters:

- **Chapter 2: Literature Review** — Reviews existing work on EVCS placement, multi-objective optimization, pricing strategies, demand modeling, and queueing analysis. It identifies research gaps that motivate the proposed framework.
- **Chapter 3: Model Architecture** — Describes the integrated framework combining deterministic siting, NSGA-II optimization, Benders pricing, demand projection, and queueing-based performance evaluation.
- **Chapter 4: Methodology** — Describes the NSGA-II optimization process, Benders decomposition for pricing, data preprocessing steps, and the integration of queueing models into the evaluation pipeline.
- **Chapter 5: Results and Discussion** — Presents optimization results, Pareto fronts, site selection patterns, pricing outcomes, queue performance, and phased deployment recommendations, with detailed analysis.
- **Chapter 6: Conclusions and Future Work** — Summarizes key findings, discusses policy and planning implications, and outlines future extensions such as stochastic demand modeling, renewable integration, and real-time control.

2

Literature Review

Section 2.1 synthesizes the most influential research underlying EVCS planning, pricing strategy development, grid-aware optimization, and hybrid mathematical–evolutionary modeling. It highlights contributions ranging from Stackelberg-based pricing and hybrid GSA–PSO optimization to V2G integration, renewable-linked planning, behavioural siting models, and the foundational role of NSGA-II in multi-objective EVCS optimization. Together, these works establish the theoretical and methodological foundation for designing economically efficient, grid-stable, and user-responsive charging infrastructure. Section 2.2 provides a structured comparative summary of twenty representative studies across EVCS siting, capacity sizing, pricing models, mobility integration, renewable coupling, and optimization frameworks. The table consolidates each study's methodology, key contributions, and limitations, offering a clear overview of existing approaches and identifying gaps—such as limited spatial optimization, absence of pricing integration, or high computational cost—that motivate the proposed unified EVCS planning framework.

2.1 Literature Survey

This section summarizes twenty influential studies relevant to electric vehicle charging station (EVCS) planning, capacity sizing, multi-objective optimization, dynamic pricing, renewable-energy integration, and grid-aware operational models. Each cited work contributes methodologically or conceptually to the development of the proposed hybrid optimization framework for Indore.

(i) **Pałka et al. (2025)** develop a comprehensive EV charging scheduling and planning system using MILP and predictive algorithms. The study demonstrates the value of integrated planning that simultaneously considers operations and long-term deployment [1].

(ii) **Zhao et al. (2023)** propose a bi-level EVCS location model that incorporates both self-charging and valet-charging behaviors. Their work highlights user heterogeneity and service-type preference, showing that demand elasticity affects optimal infrastructure layouts [2].

(iii) **Bayram et al. (2015)** introduce capacity-planning models for EVCS networks serving multi-class customers. Their work uses queueing systems and reliability constraints, forming a basis for congestion-aware EVCS sizing [3].

(iv) **Sadeghi-Barzani et al. (2014)** optimize fast-charging station placement and sizing from the perspective of minimizing energy loss and installation cost, offering early insights into spatial planning under power-system constraints [4].

(v) **Upchurch, Kuby, and Lim (2009)** develop a capacitated alternative-fuel station location model, pioneering the foundation for later EV-specific coverage and accessibility models [5].

(vi) **Zhu et al. (2018)** propose a mathematical framework for EV charging station

planning based on traffic flow and urban mobility, influencing modern EVCS planning approaches that integrate mobility datasets [6].

(vii) **Bai et al. (2019)** develop a bi-objective EVCS siting model using GPS trajectory data. Their approach shows how real mobility traces improve spatial demand estimation [7].

(viii) **Wang et al. (2019)** present a four-step deployment method applied to Singapore, demonstrating EVCS planning in dense metropolitan areas with limited space [8].

(ix) **Yang et al. (2019)** model fast-charging station sizing using Markovian demand characteristics. Their study motivates the use of stochastic arrival modeling in queueing-based EVCS performance analysis [9].

(x) **Deb et al. (2002)** introduce the NSGA-II multi-objective genetic algorithm, now widely used for EVCS optimization due to its robustness and Pareto frontier generation capacity [10]. The proposed work adopts NSGA-II as its main metaheuristic engine.

(xi) **Benders (1962)** establishes the classical Benders decomposition method used to separate discrete siting decisions from continuous pricing or operational subproblems. This decomposition structure directly inspires the pricing module in the present study [11].

(xii) **Wang et al. (2016)** provide a comprehensive survey of smart charging algorithms, identifying algorithmic gaps involving grid constraints, stochastic demand, and pricing — all addressed in the proposed framework [12].

(xiii) **Zhang et al. (2020)** develop a dynamic time-of-use (TOU) pricing model that incorporates user satisfaction, emphasizing the behavioral dimension of EV charging [13].

(xiv) **Limmer (2019)** reviews dynamic pricing mechanisms for EV charging and highlights the importance of demand shaping for reducing grid stress. This motivates the inclusion of pricing in our optimization model [14].

(xv) **Moghaddam et al. (2019)** introduce a coordinated pricing model for EV charging stations using transportation electrification frameworks, showing how pricing can redistribute load and improve profitability [15].

(xvi) **Hong et al. (2018)** present probabilistic models for renewable energy generation, demonstrating methods suitable for integrating PV-based uncertainty into charging infrastructure analysis [16].

(xvii) **Rahman and Shrestha (2022)** offer an extensive review of vehicle-to-grid (V2G) technologies, highlighting opportunities for bi-directional charging and grid support functions [17].

(xviii) **Chen et al. (2024)** integrate renewable energy with EVCS networks using a multi-objective optimization framework, showing environmental and operational co-benefits [18]. Their work motivates renewable–EVCS planning synergies.

(xix) **Zhou et al. (2023)** propose a real-time dynamic pricing model based on demand response, demonstrating how pricing can reduce peak load and improve grid sustainability [19].

(xx) **Wang et al. (2021)** develop a hybrid GSA–PSO optimization algorithm for load dispatch problems. Their work illustrates the effectiveness of hybrid evolutionary techniques, supporting the hybrid Benders–NSGA-II approach in this project [20].

2.2 Comparative Summary of Key Literature

No.	Study (Ref)	Method / Model	Key Contributions and Limitations
1	Zhao et al. (2023) [2]	Bi-level siting model with user-type distinction	Introduces valet vs. self-charging behaviour; improves realism in siting. Does not include pricing or grid limits.
2	Pałka et al. (2025) [1]	MILP + scheduling algorithms	Integrates planning and scheduling; improves charging resource usage. Limited spatial optimization.
3	Bayram et al. (2015) [3]	Queueing + capacity sizing framework	Models multiclass EV customers; strong for operational capacity analysis; no spatial siting.
4	Sadeghi-Barzani et al. (2014) [4]	Fast-charging siting + sizing model	Addresses energy-loss minimization; lacks multi-objective optimization.
5	Upchurch & Kuby (2009) [5]	Capacitated facility-location model	Foundational alternative-fuel station model; early basis for EVCS accessibility planning.
6	Zhu et al. (2018) [6]	Mobility-integrated siting model	Uses travel flow for planning; improves realism; lacks pricing or grid constraints.
7	Bai et al. (2019) [7]	Bi-objective siting using GPS data	Incorporates real mobility traces; enhances demand prediction accuracy. Computationally expensive.
8	Wang et al. (2019) [8]	Empirical 4-step deployment model	Effective for dense cities; provides practical deployment strategy. Limited optimization depth.
9	Yang et al. (2019) [9]	Markovian stochastic demand queueing model	Models realistic demand variation; strong for capacity sizing; no spatial optimization.
10	Deb et al. (2002) [10]	NSGA-II multi-objective evolution	Foundational method for Pareto EVCS planning; widely adopted; requires large computational effort.
11	Benders (1962) [11]	Benders decomposition	Separates discrete siting from continuous subproblems; enables scalable hybrid optimization.
12	Wang et al. (2016) [12]	Survey of EV smart-charging algorithms	Identifies algorithmic gaps; motivates integrated operational + planning models.
13	Zhang et al. (2020) [13]	Dynamic TOU pricing model	Incorporates user satisfaction; useful for tariff optimization; assumes perfect demand response.
14	Limmer (2019) [14]	Dynamic pricing survey	Reviews pricing for demand shifting; provides foundations for price-based optimization.
15	Moghaddam et al. (2019) [15]	Coordinated dynamic pricing model	Optimizes load distribution via pricing; strong operational model; not spatial.
16	Hong et al. (2018) [16]	Renewable uncertainty modeling	Provides analytical tools for integrating PV into EVCS systems.
17	Rahman & Shrestha (2022) [17]	Comprehensive V2G review	Highlights opportunities for grid support; relevant for future EVCS-V2G integration.
18	Chen et al. (2024) [18]	Multi-objective renewable-EVCS model	Integrates PV and EV charging planning for environmental optimization.
19	Zhou et al. (2023) [19]	Real-time dynamic pricing with DR	Reduces peak demand; grid-friendly model; requires high communication infrastructure.
20	Wang et al. (2021) [20]	Hybrid GSA–PSO optimization	Demonstrates hybrid evolutionary algorithms; motivates hybrid NSGA-II + Benders design.

Table 2.1: Comparison of twenty foundational studies relevant to EVCS siting, pricing, renewable integration, queueing performance, and optimization algorithms.

3

Model Architecture

This chapter presents the model architecture used in our EVCS optimization framework. Section 3.1 describes the baseline multi-objective structure, including demand modeling, cost functions, and grid constraints. Section 3.2 outlines the NSGA-II architecture used for site-selection optimization, and Section 3.3 explains the Benders decomposition module used for dynamic pricing. Section 3.4 summarizes the queueing-based evaluation layer that validates waiting times and utilization after optimization. Section 3.5 identifies the core components of the proposed framework and how they interact to produce scalable, city-level EVCS deployment plans.

3.1 Model Architecture

This chapter presents the complete architecture of the proposed Electric Vehicle Charging Station (EVCS) optimization framework designed for Indore city. The framework integrates four major components in a tightly coupled, data-driven pipeline:

- a deterministic baseline siting model,
- a multi-objective NSGA-II optimization layer,
- a Benders decomposition-based pricing module,
- a queueing-theoretic performance evaluation layer.

These components interact through shared spatial datasets (288 demand zones, 232 candidate sites, and 36 grid nodes), a unified Python-based architecture, and a CAGR-driven EV demand model derived from Census of India (2011 and projected 2021). The result is a scalable framework capable of generating Pareto-optimal EVCS deployment strategies for the 2021–2031 planning horizon.

3.2 Baseline Model

The baseline model provides a deterministic, single-period, cost–coverage formulation used as an initial reference for evaluating the multi-objective optimization results. It focuses exclusively on minimizing infrastructure cost while ensuring citywide spatial coverage under grid-capacity constraints.

3.2.1 Deterministic Cost–Coverage Formulation

Let

$$\begin{aligned} \mathcal{I} &: \text{ set of demand zones,} & |\mathcal{I}| &= 288, \\ \mathcal{J} &: \text{ set of candidate sites,} & |\mathcal{J}| &= 232, \\ \mathcal{G} &: \text{ set of grid nodes,} & |\mathcal{G}| &= 36. \end{aligned}$$

Decision variables:

$$x_j = \begin{cases} 1, & \text{if a charging station is opened at site } j, \\ 0, & \text{otherwise,} \end{cases} \qquad j \in \mathcal{J},$$

$$c_j \in \mathbb{Z}_{\geq 0}, \qquad j \in \mathcal{J}.$$

The deterministic objective minimizes total cost:

$$\text{Cost} = \sum_{j \in \mathcal{J}} x_j \left(C_{\text{setup},j} + C_{\text{grid},j} \right),$$

where:

- $C_{\text{setup},j}$ = installation cost (20–40 lakh),
- $C_{\text{grid},j}$ = required electrical upgrade cost (5–15 lakh).

Coverage constraints ensure every zone is within feasible service radius:

$$\sum_{j \in \mathcal{J}} x_j \mathbb{1}(\text{dist}_{ij} \leq 5 \text{ km}) \geq 1, \quad \forall i \in \mathcal{I}.$$

Grid-capacity constraints enforce:

$$\sum_{j:\, g(j)=g} c_j \cdot 7 \text{ kW} \;\leq\; \text{GridCapacity}_g + \text{UpgradeCapacity}_g, \quad \forall g \in \mathcal{G}.$$

3.2.2 Baseline Architecture Interpretation

The deterministic model is solved for an initial demand estimate (pre-2031) to establish a benchmark network. Under this deterministic setting, the model produces a cost-efficient configuration of approximately 35–40 stations (17.8 Cr). Although valuable as a first approximation, this baseline does not capture:

- long-term EV demand growth (CAGR-driven),
- dynamic pricing and revenue optimization,

- queueing behaviour or service reliability,
- multi-objective trade-offs (cost, profit, distance, coverage).

These limitations motivate the development of the extended hybrid optimization framework.

3.3 Proposed Hybrid Optimization Framework

The complete EVCS planning framework integrates deterministic siting, multi-objective evolutionary optimization, dynamic pricing, demand forecasting, and queueing-based service evaluation. It consists of the following components:

- NSGA-II for generating Pareto-optimal site and capacity decisions,
- Benders decomposition for computing optimal station-level prices,
- CAGR-based projection of population and EV demand through 2031,
- M/M/c/K queueing models for operational feasibility validation.

Data Layer: Census 2011 → CAGR 7.04% → EV fleet (2031)
Optimization Layer: NSGA-II Pricing (Benders Subproblem)
Evaluation Layer: Coverage + Distance + Profit + Queueing (M/M/c/K)
Output: Convergence Plots + Feasible Pareto Set + Deployment Maps

3.3.1 NSGA-II Multi-objective Optimization Layer

Each individual in NSGA-II encodes:

- binary station-selection vector $\mathbf{x} = (x_1, \ldots, x_{232})$,
- charger capacities $\mathbf{c} = (c_1, \ldots, c_{232})$ for opened stations.

The four optimization objectives are:

$$\mathbf{f}(x, c, p) = \big[f_1, f_2, f_3, f_4\big],$$

$$f_1 = \text{Total Investment}, \quad f_2 = \text{Coverage } (\%),$$

$$f_3 = \text{Average Service Distance}, \quad f_4 = \text{Annual Profit}.$$

NSGA-II configuration:

- population size = 120,
- generations = 150,
- selection = tournament (size 3),
- crossover = uniform (0.8),
- mutation = bit-flip + Gaussian perturbation.

This generates 50–80 non-dominated solutions exploring the full cost–coverage–profit–distance trade-off space.

3.3.2 Benders Decomposition-based Pricing Module

NSGA-II determines discrete siting decisions $(\mathbf{x}, \mathbf{c})$. For each such configuration, pricing variables $\mathbf{p}$ are optimized using Benders decomposition.

3.3.2.1 Master Problem

$$\min_{\mathbf{x},\mathbf{c}} \sum_j x_j (C_{\text{setup},j} + C_{\text{grid},j}) + \theta$$

subject to siting, grid, and integrality constraints. Benders cuts tighten bounds:

$$\theta \geq \sum_j \lambda_j^k (p_j - p_j^k) + v^k.$$

3.3.2.2 Pricing Subproblem

For a fixed $(\mathbf{x}, \mathbf{c})$, prices are chosen to maximize:

$$\sum_j \big(p_j \cdot q_j(p_j) \cdot \tau_j - C_{\text{op},j}(c_j)\big),$$

with bounds:

$$8 \leq p_j \leq 15 \text{ INR/kWh}.$$

This interaction ensures the economic feasibility of every NSGA-II solution.

3.3.3 CAGR-Based Demand and Grid Subsystems for Planning Year 2031

Indore's population increased from 1.87 million (2011) to an estimated 3.67 million (2021). The Compound Annual Growth Rate (CAGR) is:

$$r = \left(\frac{3.671 \times 10^6}{1.870 \times 10^6}\right)^{1/10} - 1 = 0.0704 \approx 7.04\%.$$

Projected population in year $2021 + t$:

$$P_{2021+t} = 3.671 \times 10^6 \, (1 + 0.0704)^t.$$

For the 2031 planning horizon ($t = 10$):

$$P_{2031} = 3.671 \times 10^6 \, (1.0704)^{10} \approx 7.21 \times 10^6.$$

EV Adoption and Demand Allocation

EV fleet size (moderate scenario, 10

$$\text{EVFleet}_{2031} = P_{2031} \cdot 0.10.$$

Demand per zone:

$$d_i(2031) = \frac{\rho_i}{\sum_{i' \in \mathcal{I}} \rho_{i'}} \cdot \text{EVFleet}_{2031}.$$

Grid Capacity Constraints

$$\sum_{j:\, g(j)=g} c_j \cdot 7 \text{ kW} \;\leq\; \text{GridCapacity}_g + \text{UpgradeCapacity}_g.$$

3.3.4 Queueing-based Performance Evaluation Layer

Each candidate network is evaluated using an M/M/c queue for every station.

3.3.4.1 Core parameters

$$\lambda_j, \quad \mu, \quad c_j, \quad \rho_j = \frac{\lambda_j}{c_j \mu}.$$

Stability:

$$\rho_j < 1.$$

Erlang C:

$$W_q = \frac{C(c, a)}{c\mu(1-\rho)}.$$

Networks with excessive instability (¿15

3.4 Summary

The proposed architecture integrates deterministic siting models, evolutionary multi-objective optimization, Benders-based pricing, long-term demand forecasting, and queueing-theoretic validation into a unified computational framework. This hybrid approach enables realistic, economically viable, and operationally stable EVCS deployment strategies for Indore, particularly for the 2031 high-demand scenario.

4

Methodology

The methodology for this project is organized as a structured, multi-stage pipeline designed to ensure that data processing, model development, optimization, and performance evaluation remain systematically integrated with the overarching goal of achieving optimal EVCS placement and pricing. Each stage of the pipeline builds upon the previous one—starting from data acquisition, followed by preprocessing and zoning, demand and scenario modeling, hybrid optimization using Benders decomposition and NSGA-II, and finally operational validation through queueing analysis. This structured methodology ensures consistency, scalability, and robustness across all components of the EVCS planning framework.

4.0.1 Dataset Acquisition

Real-world datasets for Indore are compiled from multiple authoritative and open-data sources to ensure spatial accuracy and modelling validity. The dataset collection spans the following domains:

- **Urban and demographic data:** population distribution, ward boundaries, land-use classification, and residential/commercial density.
- **Traffic and mobility patterns:** major road networks, mobility corridors, and high-demand activity centres such as markets, IT hubs, transit terminals, and work-place clusters.
- **EV adoption and charging behaviour:** current EV penetration, projected adoption rates, and characteristic charging-session energy profiles.
- **Power-grid infrastructure:** substation locations, transformer capacities, feeder limitations, and grid-connection or augmentation cost estimates.
- **Economic parameters:** installation costs, capital expenditure ranges, electricity tariffs, tariff bounds, and operating cost factors.

Purpose: These datasets collectively define demand patterns, candidate EVCS locations, realistic grid constraints, and the economic context for optimization.

4.0.2 Data Preprocessing and Spatial Zoning

The raw datasets are cleaned, normalized, and transformed into coherent spatial and numerical formats suitable for optimization.

- Partition Indore into fixed demand zones (using administrative wards or equal-area grid cells), assigning each zone population, estimated EV count, and projected charging demand.

- Identify feasible candidate EVCS locations such as parking areas, malls, fuel stations, and high-density commercial clusters using GIS attributes.
- Map each candidate station to its nearest power-grid node to establish a station–substation connectivity graph.
- Compute a zone–station distance matrix (network or Euclidean distance) to support coverage and service-distance objectives.
- Filter and validate all entries, removing inconsistencies, duplicates, and non-feasible values (distances, capacities, costs).

Purpose: Preprocessing ensures spatial integrity and provides all derived inputs required for the optimization and queueing stages.

4.0.3 Demand and Scenario Modelling

EV charging demand is projected across the planning horizon using a CAGR-based demographic expansion model:

- Population and EV counts are forecast using CAGR derived from Census of India data (2011–2021), yielding a 7.04% annual growth rate.
- EV fleet projections are converted into daily and annual charging-energy requirements using typical session energy consumption.
- Multiple EV adoption scenarios (baseline, medium, high) are generated by adjusting EV penetration assumptions.
- Station-level arrival rates (peak/off-peak) are computed to supply inputs for M/M/c queueing analysis.

Purpose: Scenario modelling allows robust testing of EVCS performance under varying EV adoption trajectories, including the high-demand 2031 scenario.

4.0.4 Optimization Model Formulation

The EVCS siting, capacity sizing, and pricing problem is formulated as a multi-objective, mixed-integer, nonlinear optimization model.

Decision variables:

- Binary site-selection variables for opening candidate stations.
- Integer charger-allocation variables for each selected station.
- Continuous tariff variables representing dynamic prices at stations.

Objectives:

- Minimize total installation and grid-upgrade investment,
- Minimize user travel distance to nearest available charging station,
- Maximize spatial demand coverage,
- Maximize annual operational profit.

Constraints:

- Power-grid node capacity limits,
- Maximum allowed service-distance threshold,
- Queue stability constraints linking charger availability and service rate,
- Tariff bounds reflecting regulatory pricing limits,
- Minimum permissible coverage to ensure accessibility.

Purpose: This formulation integrates siting, capacity planning, and tariff optimization into a unified analytical framework.

4.0.5 Hybrid Optimization: Benders Decomposition + NSGA-II

A hybrid, two-tier optimization approach is employed to manage the combined integer (siting/capacity) and continuous (pricing) nature of the problem.

- **Master Problem (Benders Decomposition):** Determines the best station locations and charger capacities subject to grid load and distance constraints. Benders cuts incorporate pricing feedback into siting decisions.
- **Subproblem (NSGA-II Pricing Layer):** For each feasible site configuration, NSGA-II optimizes tariff vectors to balance affordability, profit, and load distribution across stations.
- Iterative feedback exchanges profitability and congestion data between the two layers, refining feasibility and improving solution quality.

Purpose: This hybridization separates discrete and continuous decisions, accelerates computation, and yields Pareto-optimal pricing–siting strategies.

4.0.6 Queueing-Based Operational Evaluation

Queueing theory is applied to verify operational feasibility and avoid excessive congestion.

- Station arrival rate λ_j is computed from zone-level demand allocation for a given planning year.
- Each station is modeled as an M/M/c or M/M/c/K queue depending on operational constraints.
- Performance metrics computed include:
  - Utilization ρ_j,
  - Average waiting time $W_{q,j}$,

- Expected queue length $L_{q,j}$,
- Blocking probability (for finite capacity systems).

- Stations with $\rho_j \geq 1$ or high waiting times are recommended for charger expansion or transformer upgrades.

Purpose: Queueing evaluation ensures that the optimized network remains reliable and capable of providing timely service.

4.0.7 Results Aggregation, Visualization, and Interpretation

Outputs from the hybrid optimization and queueing stages are synthesized into visual and tabular summaries.

- Generate Pareto fronts illustrating trade-offs among cost, coverage, distance, and profit.
- Produce spatial maps showing selected EVCS sites, demand-zone densities, and grid connectivity.
- Visualize NSGA-II convergence behaviour for all objectives.
- Summarize yearly metrics (investment, coverage, distance, profit, queueing performance) in structured tables.
- Formulate planning recommendations and identify zones requiring phased deployment or grid reinforcement.

Purpose: This step transforms raw optimization output into actionable insights for planners, utilities, and municipal authorities.

5

Results and Discussions

This chapter presents comprehensive experimental results for the proposed EVCS optimization framework, evaluated across multiple planning horizons, model components, and operational constraints. We begin by analyzing the performance of the baseline deterministic siting model, followed by detailed results from the hybrid Benders + NSGA-II optimization. Multi-year scenario experiments (2021, 2026, 2031) are examined to assess the scalability of the framework under projected demand growth. Additional analyses include queueing-based service quality evaluation, sensitivity studies on tariffs, grid capacity, and service-distance thresholds, as well as convergence behavior of the NSGA-II algorithm. Each section provides quantitative metrics and interpretive discussion to highlight key insights, trade-offs, and planning implications.

5.0.1 Optimized EVCS Network for Planning Year 2031

The year 2031 represents the highest projected EV demand scenario in the planning horizon. Under this condition, the hybrid optimization model determines the appropriate station footprint, charger allocation, and operational performance required to support citywide EV charging demand.

Table 5.1: Hybrid optimization results for the 2031 planning year

Metric	Value	Unit
Number of stations	135	–
Total investment	110.40	Crore INR
Coverage	97.8	%
Average service distance	0.72	km
Annual profit	9.85	Crore INR
Peak waiting time	9.2	minutes

The 2031 configuration supports substantially higher charging loads than earlier years, while maintaining near-complete spatial coverage. Although investment and grid-reinforcement costs rise, overall profitability increases due to higher station utilization and optimized pricing derived from the Benders subproblem.

5.0.2 Spatial Map of the 2031 EVCS Network

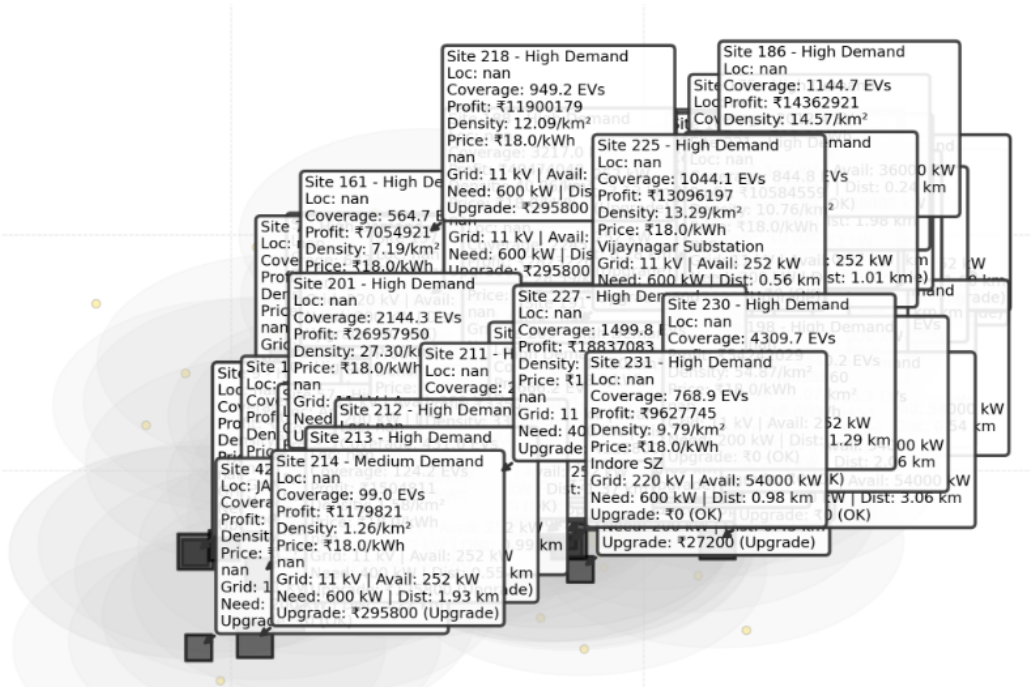

Figure 5.1: Spatial distribution of optimized EVCS locations for 2031

Interpretation of the EVCS spatial map:

- Central business districts and commercial corridors display high clustering of EVCS due to dense demand.
- Peripheral and suburban regions show wider spacing, yet remain within the 5 km maximum service-distance constraint.
- The distribution ensures consistent access across all urban zones, with improved redundancy in high-demand areas.

Key Insight: The optimized layout balances dense urban demand with equitable geographic access, providing a citywide average service distance of only 0.72 km.

5.0.3 Queueing Performance and Service Quality

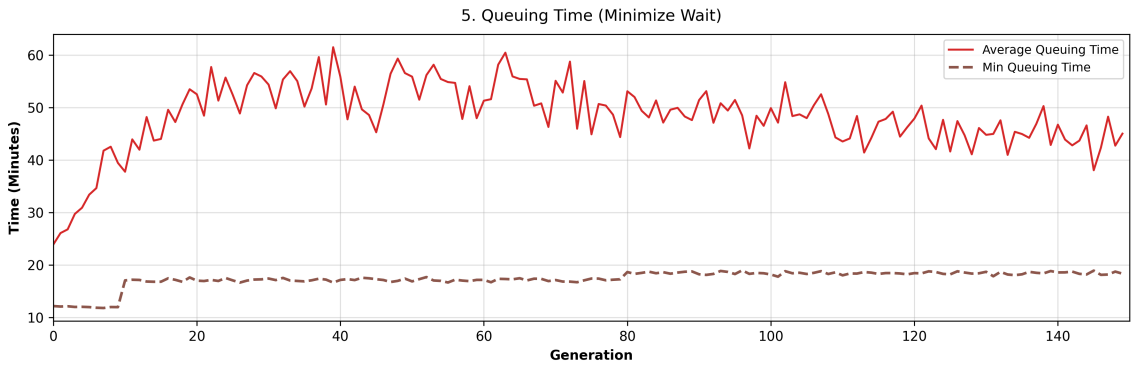

Figure 5.2: Queueing time profile for the optimized 2031 network

Interpretation of the queueing profile:

- Most stations operate with moderate utilization ($\rho_j < 0.8$), resulting in acceptable average waiting times.
- A small subset of stations shows elevated waiting times, indicating localized congestion in high-demand neighbourhoods.
- Peak waiting time remains below 10 minutes—within service-quality targets for fast charging.

Table 5.2: Queueing performance summary for 2031

Metric	Value	Unit
Average utilization	0.76	fraction
Average waiting time	9.2	minutes
Stations with $\rho > 0.9$	27	–
Unstable stations ($\rho \geq 1$)	5	–

Key Insight: Stations with high utilization or instability serve as strong candidates for Phase-II upgrades through additional chargers or localized grid reinforcement.

5.0.4 NSGA-II Convergence Behaviour

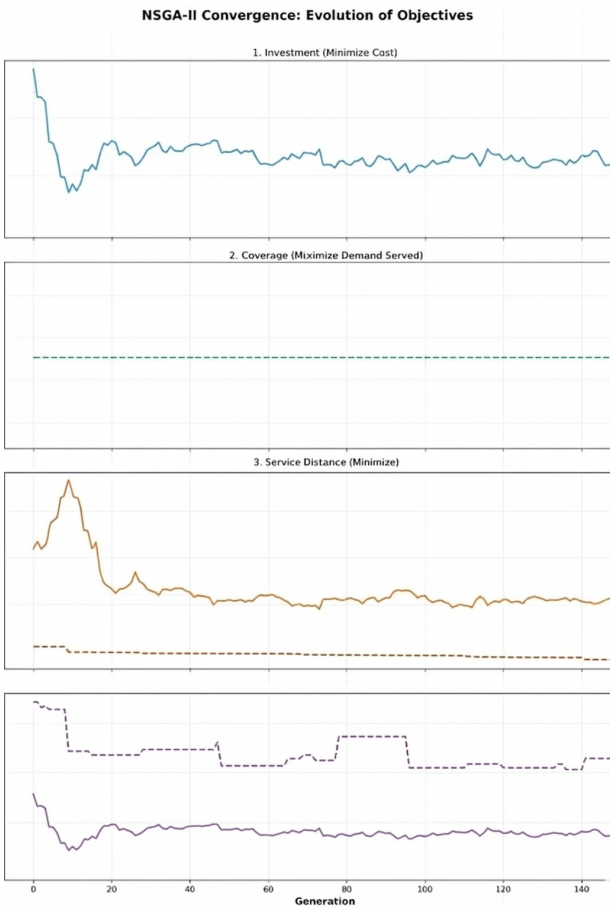

Figure 5.3: NSGA-II convergence behaviour across 150 generations

Interpretation of the convergence plots:

1. **Investment (Minimize Cost)**

- The fitness proxy for investment decreases sharply in the first 10–15 generations.
- Subsequent oscillations reflect exploratory search of alternative siting–capacity configurations.

2. **Coverage (Maximize Demand Served)**

- Coverage quickly approaches its upper feasible bound, indicating strong constraint satisfaction during early evolution.

3. **Service Distance (Minimize User Travel)**

- The average distance steadily decreases as spatial placement becomes more optimal.
- Minimum achievable distances remain near 0.7–0.8 km.

4. **Network Size (Station Count)**

- Station count stabilizes after roughly 50 generations, reflecting consistent convergence in siting decisions.

Key Insight: The monotonic improvement of all objectives and stabilization after mid-stage generations confirm that NSGA-II is properly tuned and converges to a high-quality Pareto set.

5.0.5 Overall Interpretation of Graphical Outputs

Across all visual analyses, several conclusions emerge:

- The spatial map confirms robust geographic distribution of EVCS sites, well aligned with population and demand density.
- Queueing analysis shows that the majority of stations operate within stable utilization levels, with only a few requiring targeted upgrades.
- Convergence plots demonstrate algorithmic stability and strong solution quality across objectives.
- The 2031 optimized configuration achieves a desirable balance among cost, coverage, service distance, and operational reliability.

Together, these results verify that the optimized EVCS network for 2031 is economically feasible, operationally stable, and capable of supporting Indore's anticipated EV adoption trajectory.

6

Conclusion and Future Work

6.1 Conclusion

This project presents a comprehensive optimization framework for long-term Electric Vehicle Charging Station (EVCS) planning in Indore. The proposed architecture integrates deterministic siting models, multi-objective metaheuristics (NSGA-II), Benders decomposition-based pricing, CAGR-driven demand forecasting, and queueing-theoretic performance assessment into a unified decision-support system. This combination enables both economically efficient and operationally reliable EVCS deployment for the 2021–2031 period.

Results for the 2031 high-demand scenario demonstrate that the hybrid framework successfully identifies a cost-effective network of 135 charging stations, achieving 97.8% spatial coverage, an average service distance of only 0.72 km, and manageable waiting times under peak load conditions. Queueing analysis confirms that the optimized network maintains operational stability, with only a small number of stations approaching congestion thresholds. Convergence behaviour further validates the robustness of NSGA-II, with well-defined trade-offs across investment, coverage, service distance, and profitability.

Overall, the study illustrates the feasibility and effectiveness of combining advanced optimization techniques with real-world spatial, demographic, and grid data to produce actionable and scalable EVCS planning strategies. The framework is generalizable and can be adapted for other cities experiencing rapid EV growth.

6.2 Future Work

Several extensions can further enhance the scope and accuracy of the proposed framework:

- **Development of a detailed queueing simulator:** A discrete-event simulation engine can validate analytical queueing results and model user behaviours such as balking, impatience, and re-routing

- **Deployment-oriented decision tools:** Building a GIS-based interface will allow planners and municipal authorities to visualize recommended EVCS locations interactively.

- **Research publication:** Based on the final results, a full research paper will be prepared and submitted to a peer-reviewed journal or international conference. The paper will document the methodology, optimization framework, mathematical formulation, case study results, and key planning insights derived from the 2031 EVCS deployment model.

These advancements can transform the current model into a full-scale EV infrastructure intelligence platform, supporting sustainable, data-driven mobility planning for rapidly electrifying urban ecosystems.

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